

Pediatric Dosing Band Generator

First Principles, Method, and Evidence Base

Clinical Informatics · Pharmacy · Pediatrics

Purpose

Weight-based pediatric dosing requires translating a target dose per kilogram into a single dispensable quantity that is safe across a range of patient weights. Doing this manually for every drug formulation — accommodating the full span from the 300-gram premature infant to the bariatric adolescent — is time-intensive, inconsistent, and error-prone. This application automates the generation of standardized weight-band dosing tables, one per drug formulation, for use by pharmacists and prescribers building order sets, formulary entries, and EHR decision support tools.

The output is a complete, printable table that can be handed directly to an EHR implementation team without additional manual smoothing. The application requires only four inputs: drug formulation, target dose per kilogram, maximum dose, and declared variance tolerance.

The Mathematics of Weight Bands

Every dosing band represents a clinical compromise: a fixed dispensable dose applied across a range of patient weights. The core mathematical relationship governing band construction is:

$$WtL = D \div (DT \times (1 + V)) \qquad WtH = D \div (DT \times (1 - V))$$

Where **D** is the band dose, **DT** is the target dose per kilogram, **V** is the declared variance fraction, **WtL** is the lightest patient weight the band serves, and **WtH** is the heaviest. Variance is expressed from each patient's perspective: the lightest patient receives exactly +V above their ideal dose; the heaviest receives exactly -V below theirs. The symmetry is proven algebraically, not approximated.

A critical geometric consequence is that the ratio WtH/WtL is constant across the entire weight range for a given tolerance — it depends only on V, not on the drug, dose, or formulation. This ratio determines how many bands are needed to span the full weight range and explains why band density is naturally high at low weights (narrow absolute width) and low at high weights (wide absolute width).

Variance Tolerance	WtH / WtL Ratio	Band Width at 10 kg	Typical Clinical Context
± 5%	1.105×	10.0–11.1 kg	Narrow therapeutic index
± 10%	1.222×	10.0–12.2 kg	Moderate safety margin
± 15%	1.353×	10.0–13.5 kg	Broad therapeutic index
± 20%	1.500×	10.0–15.0 kg	Wide-index drugs, standard

Band Construction: The Sequential Filter

Tables are built by the sequential filter algorithm, which walks through the complete set of physically dispensable doses for a formulation in ascending order and applies a single retention test: retain a dose if and only if its WtL is at or beyond the previous retained dose's WtH. This ensures:

- No weight gap — every patient weight maps to exactly one band
- No overlap — band boundaries are exclusive upper and inclusive lower
- Minimum clinically necessary rows — doses that are redundant (their weight range already served by the previous dose within tolerance) are dropped automatically
- Natural density gradient — fine-grained bands at low weights, coarser bands at high weights, without any explicit row-count target

The algorithm does not attempt to minimize rows or maximize variance utilization. Its goal is clinically necessary and communicable doses with safe variances — a distinction that preserves intermediate doses that a purely mathematical optimizer would eliminate.

Syringe-First Candidate Generation

Every candidate dose presented to the sequential filter must be physically dispensable on an active syringe without rounding. The algorithm generates candidate doses directly from the legal graduation volumes of the active syringe set, not from a mathematical series that is approximated afterward. This eliminates the need for post-selection snapping and guarantees that every row in the table reflects a dose a nurse can actually draw up.

Syringe assignment is volume-range-based, not concentration-based:

- **< 1.5 mL:** 1 mL syringe (0.05 mL standard; 0.01 mL neonatal)
- **1.5–3.0 mL:** 3 mL syringe (0.1 mL graduations)
- **3.0–5.0 mL:** 5 mL syringe (0.2 mL + X.5 mL marks)
- **5.0–10.0 mL:** 10 mL syringe (0.2 mL + X.5 mL marks)
- **> 10 mL:** 10 mL multiple draws — logistics deferred to pharmacist and nurse

Within each syringe's domain, a simplicity tier hierarchy determines which graduation is preferred when multiple options satisfy the tolerance criterion: whole mL (tier 0) → X.5 mL (tier 1) → 0.2 mL steps (tier 2) → 0.1 mL steps (tier 3) → 0.05 mL steps (tier 4) → 0.01 mL steps (tier 5). The algorithm always selects the simplest readable graduation that serves the weight band, falling to finer steps only when the band geometry requires it.

Selected References

1. Jordan MA et al. Development of Guidelines for Accurate Measurement of Small Volume Parenteral Products Using Syringes. *Hospital Pharmacy*. 2021;56(3):165–171.
2. Meyers RS. The Past, Present, and Future of Oral Dosage Forms for Children. *J Pediatr Pharmacol Ther*. 2024;29(1):22–31.
3. Yin HS et al. Liquid Medication Errors and Dosing Tools: A Randomized Controlled Experiment. *Pediatrics*. 2016;138(4):e20160357.
4. U.S. Food and Drug Administration. Tablet Scoring: Nomenclature, Labeling, and Data for Evaluation. Guidance for Industry. March 2013.
5. Chaudhri S et al. Does splitting a tablet obtain an accurate dose? Systematic review and meta-analysis. *J Pharmacy Practice and Research*. 2022;52(3):210–220.
6. American Academy of Pediatrics Committee on Drugs. Metric Units and the Preferred Dosing of Orally Administered Liquid Medications. *Pediatrics*. 2015;135(4):784–787.
7. ISMP. List of Oral Dosage Forms That Should Not Be Crushed. Institute for Safe Medication Practices; updated periodically.

Delivery Device Standards and Dose Selection

Syringes, Quantum Logic, Solid Formulations, and Weight Precision

Oral Syringes: Safety Standards and Syringe Selection

The accuracy of any liquid dose is bounded by the physical precision of the delivery device. Oral syringes are the recognized gold standard for pediatric liquid medication delivery, consistently outperforming dosing cups in accuracy studies. Device accuracy degrades when a syringe is filled below 20% of its labeled capacity — the documented hard floor below which USP $\pm 10\%$ tolerance cannot be reliably maintained. The clinical target is 50% fill or greater, achieving $\pm 5\%$ device accuracy. Syringe assignment rules implement this directly: the smallest syringe that holds the dose at 50% fill or better is preferred.

The 1.0–1.5 mL overlap zone allows either the 1 mL or 3 mL syringe. The 1 mL syringe is preferred for precision; the 3 mL syringe is acceptable when the 1 mL syringe is unavailable. For doses below 0.3 mL in the neonatal setting, the 0.01 mL syringe provides the finest achievable precision and is the required device.

Syringe	Graduation	Dose Range	Optimal Fill	Notes
1 mL	0.05 mL	< 1.5 mL	$\geq 20\%$ capacity	NICU: use 0.01 mL syringe
3 mL	0.1 mL	1.5 – 3.0 mL	≥ 1.5 mL	Workhorse for infant doses
5 mL	0.2 mL + X.5	3.0 – 5.0 mL	≥ 3.0 mL	X.5 mL marks reliably readable
10 mL	0.2 mL + X.5	5.0 – 10.0 mL	≥ 5.0 mL	> 10 mL: logistics to pharmacist
Special 5 mL	1.25 mL steps	1.25 – 5.0 mL	N/A	Acetaminophen only; teaspoon legacy

Formulation Quantum and Forced Aliquots

Every liquid suspension has a natural dose quantum that emerges from the intersection of its concentration and syringe physics. The quantum is the coarsest graduation on an active syringe that produces a clinically communicable dose when multiplied by the concentration. For prednisolone at 3 mg/mL, a 0.5 mL graduation produces 1.5 mg — the natural half-quantum. A 1.0 mL graduation produces 3 mg — the primary quantum. Both are clinically communicable, so both are retained as candidate doses.

Beyond the quantum, five syringe landmark volumes are always forced into every liquid table as mandatory row anchors, regardless of what the sequential filter would select: 1 mL, 2.5 mL, 5 mL, 7.5 mL, and 10 mL (subject to the maximum dose ceiling). These correspond to the standard reference volumes on oral syringes and the natural mental-math checkpoints built into every suspension label. As demonstrated by Table 1 of Meyers 2024, drugs formulated as “x mg/5 mL” produce clean 2.5/5/7.5/10 mL doses at typical pediatric weights — a design choice by manufacturers that the table algorithm must honor and surface explicitly.

For the neonatal zone (weight < 2.5 kg), the simplicity preference is suspended entirely. Every step at the finest active syringe graduation is retained, because a 0.05 mL difference at 3 mg/mL represents 0.15 mg — a clinically meaningful distinction for a patient who may weigh 400 grams. Above 2.5 kg the tier-first simplicity preference resumes.

Weight Precision and Band Boundary Convention

Weight bands are displayed using a three-tier precision system that matches clinical measurement practice:

- < 5 kg:** hundredths of a kilogram (e.g. 2.73 kg) — NICU and neonatal range where 50-gram differences are clinically meaningful
- 5–20 kg:** tenths of a kilogram (e.g. 12.5 kg) — infant and toddler range
- ≥ 20 kg:** integer kilograms (e.g. 34 kg) — school-age and adolescent range

Band boundaries use an inclusive lower / exclusive upper convention throughout: a patient at the displayed lower bound belongs to that band; a patient at the displayed upper bound belongs to the next band. This eliminates ambiguity for patients at boundary weights. Patient weights may be entered to thousandths of a kilogram for calculation; display precision is applied only to the band boundaries in the output table. The minimum weight in any table is 0.300 kg — below this, oral dosing of the drugs covered by this application is not clinically feasible.

Solid Formulations: Divisibility Rules

Tablet divisibility is a regulatory property, not a physical inference from the presence of a score line. The FDA's 2013 Tablet Scoring Guidance distinguishes functional scoring — where the manufacturer has provided stability, content uniformity, and splitability data supporting an approved labeling claim — from decorative or coincidental score marks. Only functionally scored, immediate-release tablets should be assumed to split reliably.

The algorithm constructs solid formulation tables using a strict tier hierarchy: whole-tablet doses anchor the table, half-tablet steps fill inter-anchor gaps (if the formulation is half-splittable), and quarter-tablet steps fill any remaining gaps (if the formulation is quarter-splittable). This order is absolute. Volume must always increase monotonically across rows. Transition zones between whole-tablet anchors where no half or quarter step fits cleanly are absorbed into the adjacent band that produces the smaller maximum variance exceedance, with the resulting actual variance reported honestly. The ISMP Do Not Crush designation extends by convention to Do Not Split unless the prescribing information explicitly states otherwise.

For the current prototype, splittability is declared as a per-formulation property in the application with UI checkboxes. The production implementation should encode splittability as a database field populated from FDA functional scoring designations and the ISMP Do Not Crush list.

This document describes the methodological and evidence basis for the Pediatric Dosing Band Generator. Tables produced by the application are intended for pharmacy verification and EHR implementation use only, and do not constitute independent clinical recommendations.